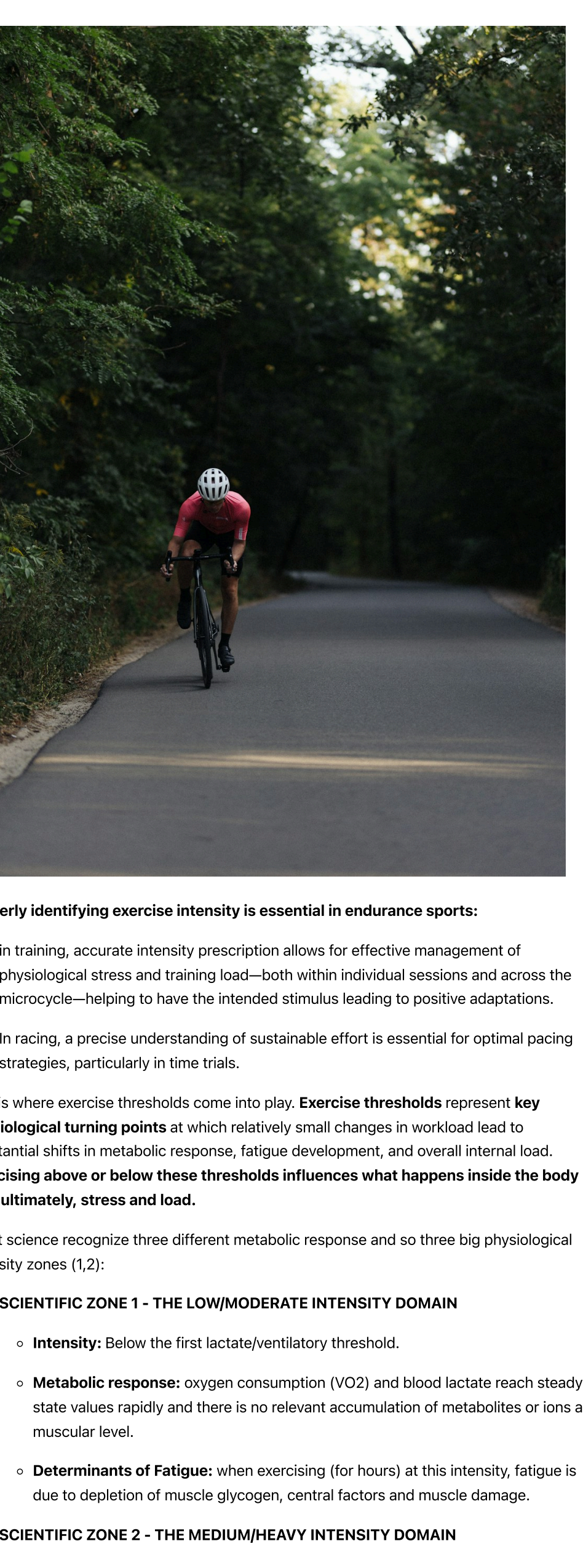
139. CRITICAL POWER: THE GOLD STANDARD FOR THE MAXIMUM METABOLIC STEADY STATE



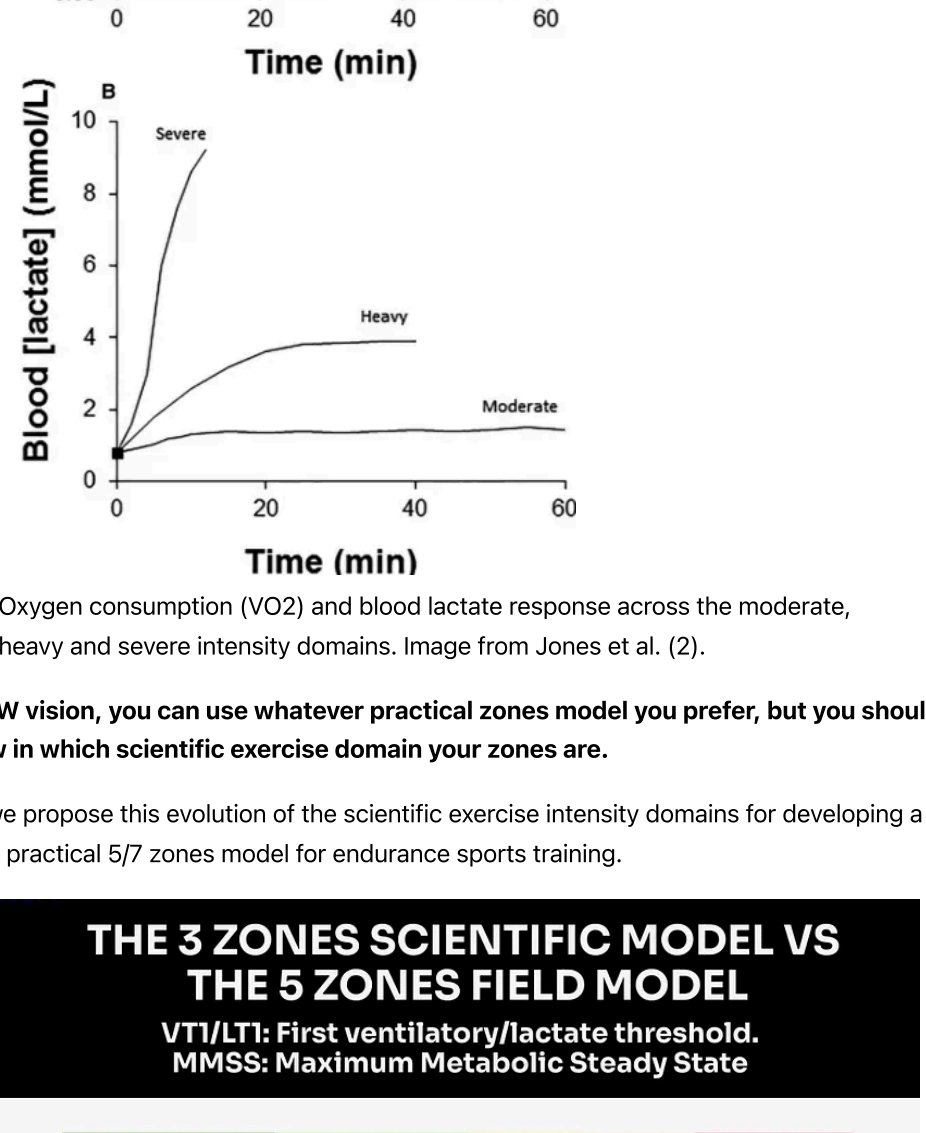
Properly identifying exercise intensity is essential in endurance sports:

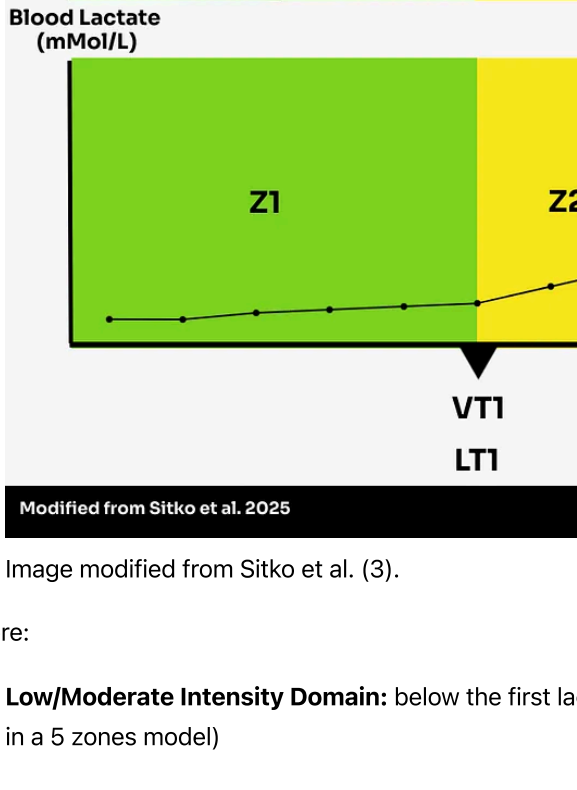
- in training, accurate intensity prescription allows for effective management of physiological stress and training load—both within individual sessions and across the microcycle—helping to have the intended stimulus leading to positive adaptations.
- In racing, a precise understanding of sustainable effort is essential for optimal pacing strategies, particularly in time trials.

This is where exercise thresholds come into play. **Exercise thresholds** represent **key physiological turning points** at which relatively small changes in workload lead to substantial shifts in metabolic response, fatigue development, and overall internal load. **Exercising above or below these thresholds influences what happens inside the body and, ultimately, stress and load.**

Sport science recognize three different metabolic response and so three big physiological intensity zones (1,2):

- **SCIENTIFIC ZONE 1 - THE LOW/MODERATE INTENSITY DOMAIN**
  - **Intensity:** Below the first lactate/ventilatory threshold.
  - **Metabolic response:** oxygen consumption (VO2) and blood lactate reach steady-state values rapidly and there is no relevant accumulation of metabolites or ions at muscular level.
  - **Determinants of Fatigue:** when exercising (for hours) at this intensity, fatigue is due to depletion of muscle glycogen, central factors and muscle damage.
- **SCIENTIFIC ZONE 2 - THE MEDIUM/HEAVY INTENSITY DOMAIN**
  - **Intensity:** between the first lactate/ventilatory threshold and the *maximum metabolic steady state* (MMSS, the maximum exercise intensity at which a steady state in oxygen consumption and muscle metabolites is still achievable).
  - **Metabolic response:** steady-state values for VO_2 and blood lactate are delayed, but still reached.
  - **Determinants of fatigue:** At this intensity, fatigue is a mix of metabolic/ionic muscular perturbation and muscle glycogen depletion.
- **SCIENTIFIC ZONE 3 - THE HIGH/SEVERE INTENSITY DOMAIN**
  - **Intensity**: Above the Maximum Metabolic Steady State**.**
  - **Metabolic response:** at this exercise intensity, even when performing a constant power bout, a steady state physiological can't be reached with oxygen consumption and blood lactate continuing to increase until VO2max and the limit of tolerance are then shortly reached, and power output can't be maintained anymore. This occurs upon reaching a critical intramuscular metabolic and ionic environment: reduced phosphocreatine availability, decreased potassium concentration, and increased hydrogen ions accumulation.
  - **Determinants of fatigue:** the critical intramuscular metabolic and ionic environment described above (linked to phosphocreatine, potassium and hydrogen ions).





Oxygen consumption (VO2) and blood lactate response across the moderate, heavy and severe intensity domains. Image from Jones et al. (2).

In KIW vision, you can use whatever practical zones model you prefer, but you should know in which scientific exercise domain your zones are.

So, we propose this evolution of the scientific exercise intensity domains for developing a more practical 5/7 zones model for endurance sports training.

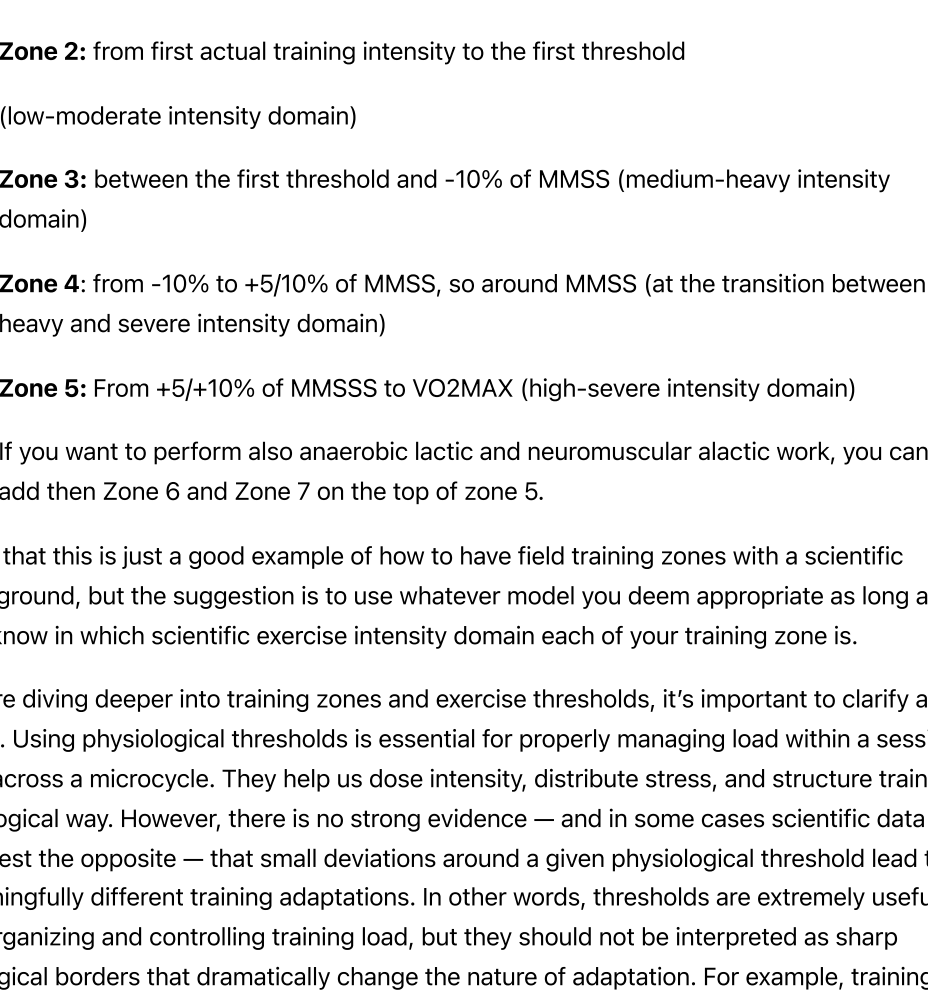


Image modified from Sitko et al. (3).

Where:

- **Low/Moderate Intensity Domain:** below the first lactate/ventilatory threshold (zone 1-2 in a 5 zones model)
- **Medium/Heavy Intensity Domain**: between the first threshold and MMSS (zone 3-low zone 4 in a 5 zones model)
- **High/Severe Intensity Domain**: above MMSS (high zone 4-zone 5 in a 5 zones model)

If you want to describe from the 5 zones model point of view:

- **Zone 1:** active recovery (low-moderate intensity domain)
- **Zone 2:** from first actual training intensity to the first threshold

  (low-moderate intensity domain)
- **Zone 3:** between the first threshold and -10% of MMSS (medium-heavy intensity domain)
- **Zone 4**: from -10% to +5/10% of MMSS, so around MMSS (at the transition between heavy and severe intensity domain)
- **Zone 5:** From +5/+10% of MMSSS to VO2MAX (high-severe intensity domain)

  If you want to perform also anaerobic lactic and neuromuscular alactic work, you can add then Zone 6 and Zone 7 on the top of zone 5.

Note that this is just a good example of how to have field training zones with a scientific background, but the suggestion is to use whatever model you deem appropriate as long as you know in which scientific exercise intensity domain each of your training zone is.

Before diving deeper into training zones and exercise thresholds, it's important to clarify a key point. Using physiological thresholds is essential for properly managing load within a session and across a microcycle. They help us dose intensity, distribute stress, and structure training in a logical way. However, there is no strong evidence — and in some cases scientific data suggest the opposite — that small deviations around a given physiological threshold lead to meaningfully different training adaptations. In other words, thresholds are extremely useful for organizing and controlling training load, but they should not be interpreted as sharp biological borders that dramatically change the nature of adaptation. For example, training slightly below or slightly above maximal metabolic steady state (MMSS) will likely stimulate very similar aerobic adaptations. The same principle applies to work performed around the first threshold: small differences in intensity do not result in completely different physiological outcomes.

Use thresholds to manage stress and plan your microcycles and interval sessions effectively — not to fragment training adaptations into different categories.

Based on what we have discussed so far, it should now be clear that identifying training intensity zones essentially comes down to defining two key physiological benchmarks:

- **The First Lactate/Ventilatory Threshold:** the boundary between the moderate- and heavy-intensity domains.
- **The Maximum Metabolic Steady State (MMSS): which marks the transition between the heavy- and severe-intensity domains. MMSS can be defined as *"the bioenergetic threshold above which fatigue development is accelerated and both systemic and muscular homeostasis can no longer be maintained."* (2)**

From a practical standpoint, these two boundaries are sufficient to structure intensity in a physiologically meaningful way.

In this KIW Research Note, we see what the gold-standard MMSS parameter should be, and why.

Have a good read!

There are two possible options to estimate the Maximum Metabolic Steady State:

1. LACTATE MEASUREMENTS
2. CRITICAL POWER

Blood lactate is the most widely used physiological marker for estimating the maximal metabolic steady state. Why lactate? Because **when the oxidative (aerobic) pathways are unable to meet the energy demands of the working muscle, pyruvate is increasingly converted into lactate rather than entering aerobic metabolism.** This shift is accompanied by metabolic disturbances, including an accumulation of hydrogen ions (H^+), which contribute to muscle fatigue and a decline in performance.

There are two options to use lactate measurements to estimate the MMSS:

- THE MAXIMUM LACTATE STEADY STATE (MLSS, note the 'L' instead of the 'M')
- THE SECOND LACTATE THRESHOLD identified from an incremental step test
- **Definition:** MLSS is defined as the highest power output at which the increase in blood lactate concentration is less than 1 mmol·L⁻¹ between the 10th and 30th minute of constant-load exercise. (3) It represents the maximal intensity at which a dynamic equilibrium still exists between lactate appearance and lactate clearance in the bloodstream. (2)
- **Testing Protocol:** a series of 30-minute constant-power trials (typically 3–5), each performed on separate days.

  During each trial, blood lactate is measured at rest and every 5 minutes throughout the exercise bout. MLSS is identified as the highest power output at which blood lactate increases by less than 1 mmol·L⁻¹ between minute 10 and minute 30. (4)

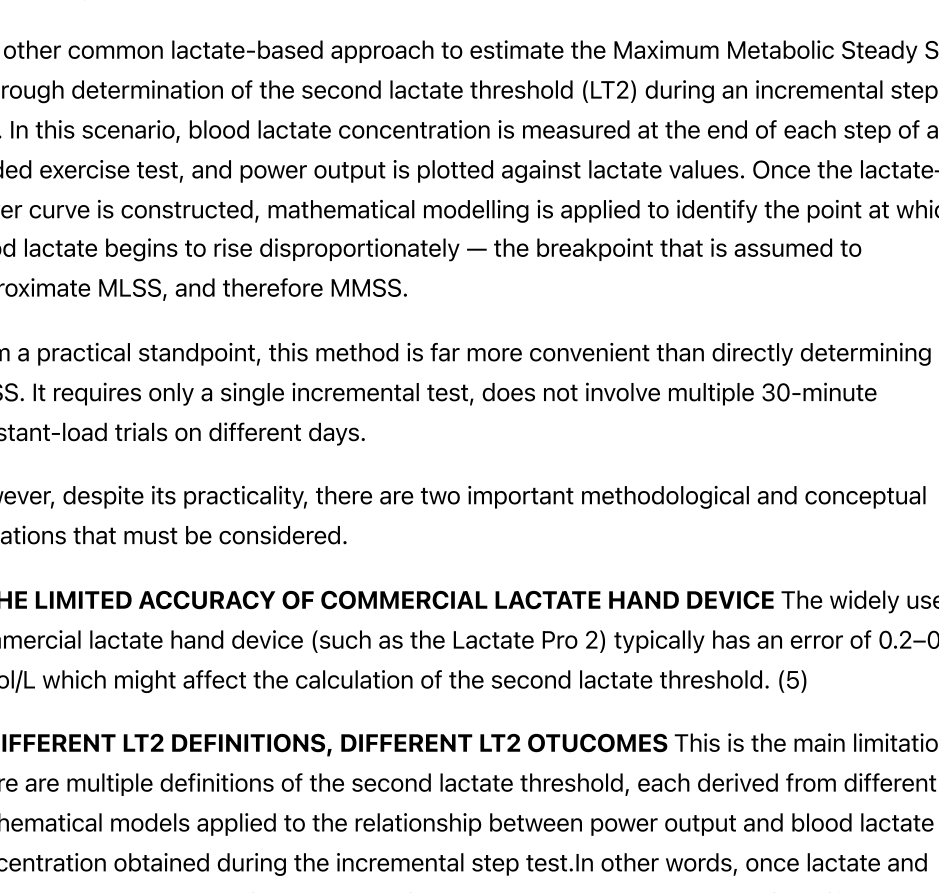


Image from Jones et al. (2)

- **Is MLSS a good estimate of MMSS?**

  Yes — provided that strict methodological criteria are followed, MLSS can offer a valid estimate of MMSS.

  However, **from a practical perspective, we do not recommend using MLSS in applied settings.**

  The reason is simple: **it is highly impractical.** A proper MLSS determination typically requires 4–6 separate 30-minute constant-power trials (with a minimum of three), each performed on different days. This means several days of testing, repeated prolonged efforts at near-threshold intensity, and a large number of lactate measurements.

  For this reason, MLSS remains largely a research tool. It is not by chance that athletes and coaches rarely — if ever — use MLSS directly to determine MMSS or to define training zones.

The other common lactate-based approach to estimate the Maximum Metabolic Steady State is through determination of the second lactate threshold (LT2) during an incremental step test. In this scenario, blood lactate concentration is measured at the end of each step of a graded exercise test, and power output is plotted against lactate values. Once the lactate–power curve is constructed, mathematical modelling is applied to identify the point at which blood lactate begins to rise disproportionately — the breakpoint that is assumed to approximate MLSS, and therefore MMSS.

From a practical standpoint, this method is far more convenient than directly determining MLSS. It requires only a single incremental test, does not involve multiple 30-minute constant-load trials on different days.

However, despite its practicality, there are two important methodological and conceptual limitations that must be considered.

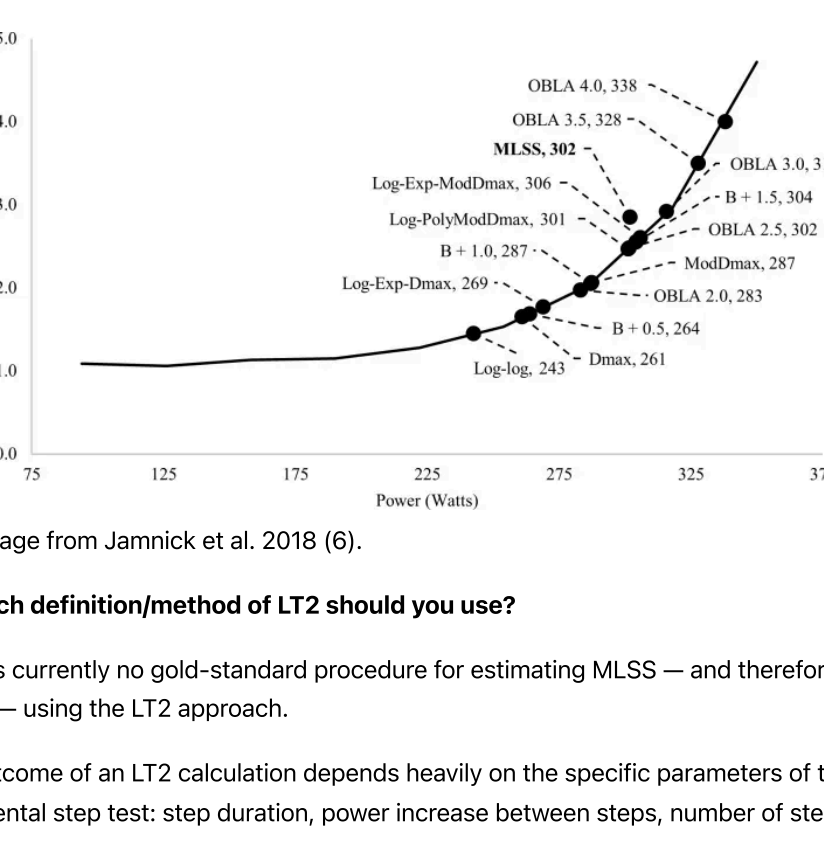
1. THE LIMITED ACCURACY OF COMMERCIAL LACTATE HAND DEVICE The widely used commercial lactate hand device (such as the Lactate Pro 2) typically has an error of 0.2–0.4 mmol/L which might affect the calculation of the second lactate threshold. (5)

2. DIFFERENT LT2 DEFINITIONS, DIFFERENT LT2 OTUCOMES This is the main limitation. There are multiple definitions of the second lactate threshold, each derived from different mathematical models applied to the relationship between power output and blood lactate concentration obtained during the incremental step test.In other words, once lactate and power data are plotted, various computational approaches can be used to identify the LT2 breakpoint.

Below are some of the most popular methods:

**OBLA: 'Onset of Blood Lactate Accumulation'*

- OBLA 3.0 mmol/L: power at a fixed blood lactate concentration of 3 mmol/L
- OBLA 3.5 mmol/L: power at a fixed blood lactate concentration of 3.5 mmol/L
- OBLA 4.0 mmol/L: power at a fixed blood lactate concentration of 4 mmol/L
- Dmax: the maximum perpendicular distance between the lactate curve and a straight line drawn connecting the first and last lactate point values



- Many different evolution of the Dmax method, which use different mathematical modelling, such as:
  - $ModD_{max}$
  - $Exp\text{-}D_{max}$
  - Log-Exp-ModDmax
  - Log-Poly-ModDmax

  But here is the thing: **different definitions of the second lactate threshold yield different power outputs for LT2.**

  In other words, the threshold you identify depends on the method you choose. The physiology of the athlete does not change — the mathematical model does.

  Below is a self-explanatory example from a single athlete tested with the same incremental protocol, but analysed using different LT2 calculation methods:

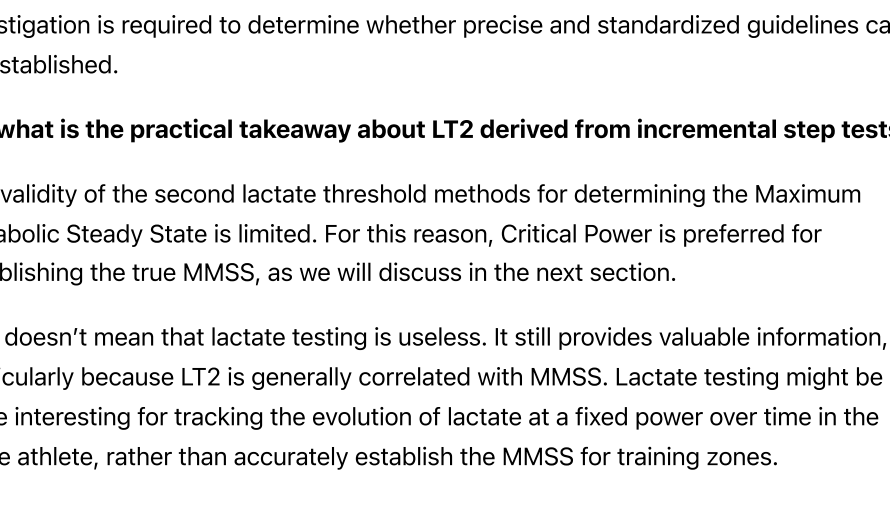


Image from Jamnick et al. 2018 (6).

So which definition/method of LT2 should you use?

There is currently no gold-standard procedure for estimating MLSS — and therefore MMSS — using the LT2 approach.

The outcome of an LT2 calculation depends heavily on the specific parameters of the incremental step test: step duration, power increase between steps, number of steps, and starting intensity. The same method — for example, Dmax — can yield different LT2 values depending on how these parameters are set. More importantly, **research has yet to consistently validate any specific test protocol for a given LT2 definition as a reliable means of accurately estimating MLSS.**

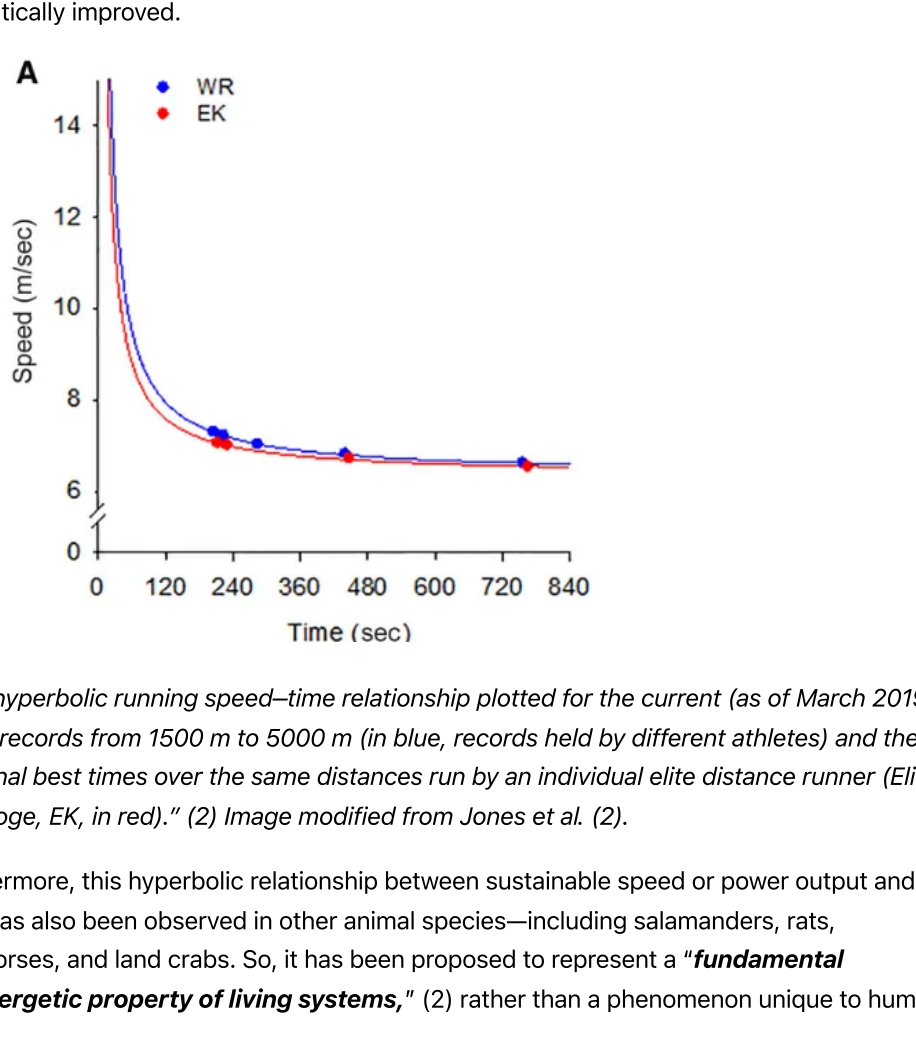
For example, a comprehensive study (6) comparing several LT2 definitions and incremental test protocols found that 4-minute step durations using the Log-Poly-$ModD_{max}$ method provided the most accurate estimate of MLSS. This is counterintuitive, as longer steps (6–10 minutes) are generally assumed to improve accuracy by allowing the physiological steady state to be reached. Nonetheless, these findings highlight that estimating MLSS/MMSS from LT2 remains an evolving area of research, and further investigation is required to determine whether precise and standardized guidelines can be established.

So, what is the practical takeaway about LT2 derived from incremental step tests?

The validity of the second lactate threshold methods for determining the Maximum Metabolic Steady State is limited. For this reason, Critical Power is preferred for establishing the true MMSS, as we will discuss in the next section.

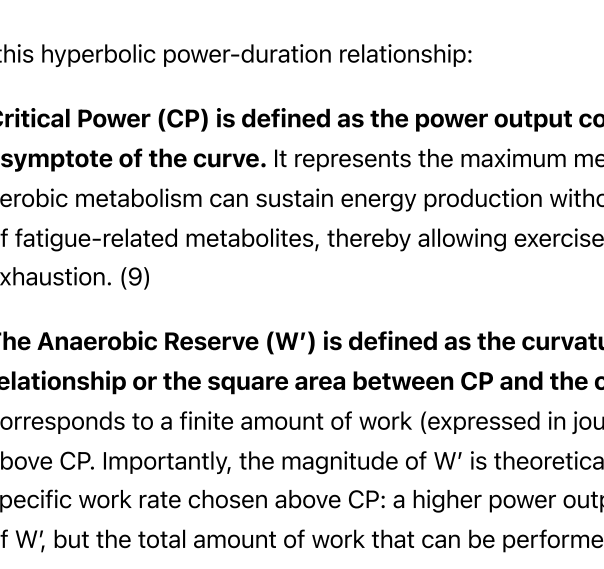
This doesn't mean that lactate testing is useless. It still provides valuable information, particularly because LT2 is generally correlated with MMSS. Lactate testing might be more interesting for tracking the evolution of lactate at a fixed power over time in the same athlete, rather than accurately establish the MMSS for training zones.

A hyperbolic relationship between speed (or power output) and the duration for which that speed or power can be sustained was first described by the legendary British exercise physiologist **Sir Archibald Hill**. In a seminal paper published in *Nature* in 1925 (!), Sir Hill identified this pattern by plotting the speed–time relationship derived from world record performances in running, swimming, and rowing (both in men and women). (7)



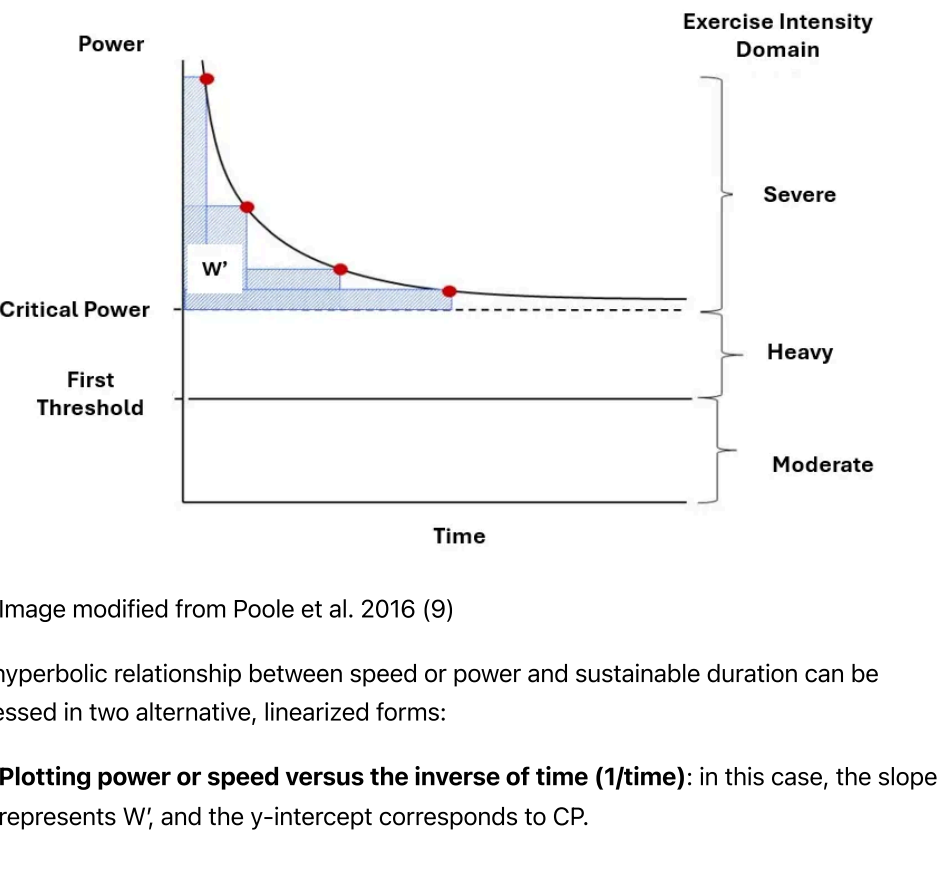
Photograph of Sir Archibald Hill alongside his original world-record hyperbolic performance relationships. Images adapted from Burnley et al. (8)

A century later, this hyperbolic relationship continues to be supported by a vast body of scientific evidence and has proven resilient even as human performance standards have dramatically improved.



"The hyperbolic running speed–time relationship plotted for the current (as of March 2019) world records from 1500 m to 5000 m (in blue, records held by different athletes) and the personal best times over the same distances run by an individual elite distance runner (Eliud Kipchoge, EK, in red)." (2) Image modified from Jones et al. (2).

Furthermore, this hyperbolic relationship between sustainable speed or power output and time has also been observed in other animal species—including salamanders, rats, racehorses, and land crabs. So, it has been proposed to represent a ***"fundamental bioenergetic property of living systems,"*** (2) rather than a phenomenon unique to humans.



From this hyperbolic power-duration relationship:

- **Critical Power (CP) is defined as the power output corresponding to the horizontal asymptote of the curve.** It represents the maximum metabolic steady state at which aerobic metabolism can sustain energy production without the continuous accumulation of fatigue-related metabolites, thereby allowing exercise to be maintained without rapid exhaustion. (9)
- **The Anaerobic Reserve (W′) is defined as the curvature constant of the relationship or the square area between CP and the curve at any point**. It corresponds to a finite amount of work (expressed in joules) that can be performed above CP. Importantly, the magnitude of W′ is theoretically fixed and independent of the specific work rate chosen above CP: a higher power output results in a faster depletion of W′, but the total amount of work that can be performed above CP remains the same. (9) It seems to represent the anaerobic capacity of the athlete.

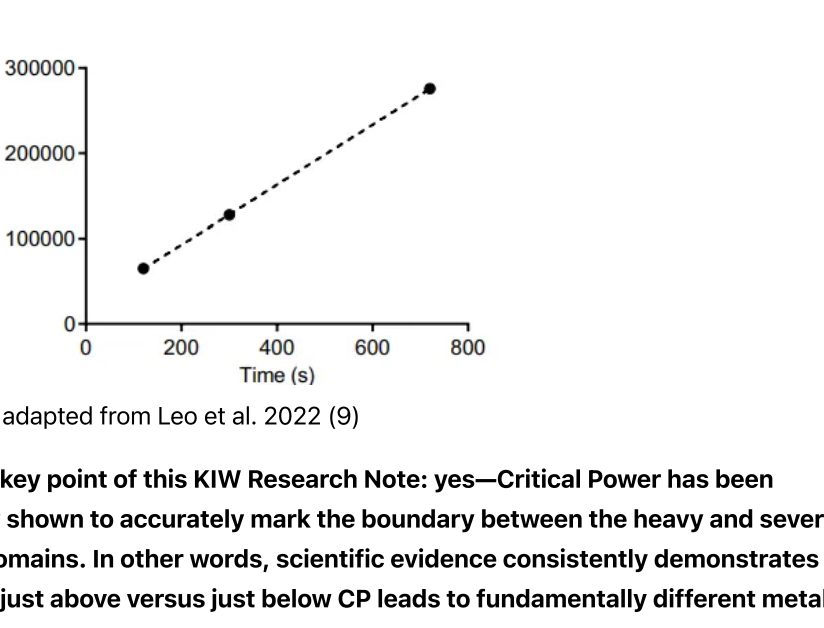


Image modified from Poole et al. 2016 (9)

The hyperbolic relationship between speed or power and sustainable duration can be expressed in two alternative, linearized forms:

- **Plotting power or speed versus the inverse of time (1/time)**: in this case, the slope represents W′, and the y-intercept corresponds to CP.
- **Plotting total work versus time**: in this representation, the slope of the resulting line corresponds to Critical Power (CP), while the y-intercept corresponds to W′, the finite work capacity above CP.

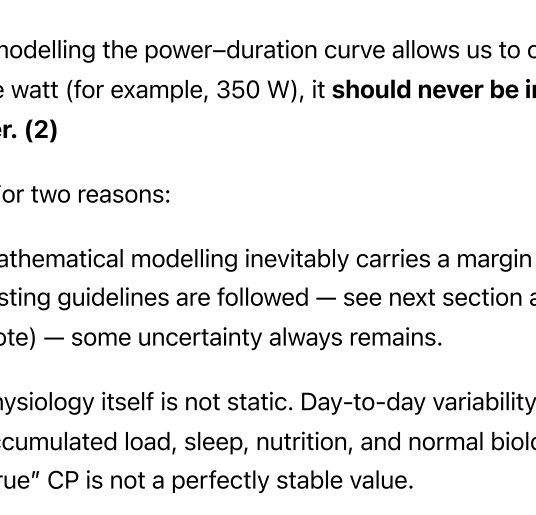


Image adapted from Leo et al. 2022 (9)

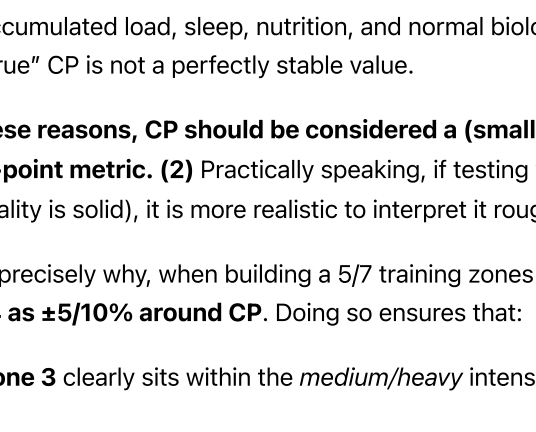


Image adapted from Leo et al. 2022 (9)

This is the key point of this KIW Research Note: yes—Critical Power has been repeatedly shown to accurately mark the boundary between the heavy and severe exercise domains. In other words, scientific evidence consistently demonstrates that exercising just above versus just below CP leads to fundamentally different metabolic responses and rates of fatigue development, confirming that CP reliably represents the Maximum Metabolic Steady State. (11)

Specifically, some of the most relevant studies investigating metabolic responses at intensities just above and just below CP have found that when exercising slightly below or at Critical Power (CP), a steady state—even if delayed (after 2-3min)—is reached for both oxygen consumption and blood lactate, and no signs of fatigue appear after 20–30 minutes. In contrast, exercising just 5-10% above CP elicits a markedly different metabolic response typical of the severe/high-intensity domain: oxygen consumption, blood lactate, intramuscular inorganic phosphate and hydrogen ions rise continuously, until VO₂max is reached, after which task failure occurs (approximately in 15-20 minutes (12,13).

While modelling the power–duration curve allows us to calculate **Critical Power** (CP) down to a single watt (for example, 350 W), it **should never be interpreted as an absolute, fixed number. (2)**

Why? For two reasons:

1. Mathematical modelling inevitably carries a margin of error. Even when best-practice testing guidelines are followed — see next section and the upcoming KIW Research Note) — some uncertainty always remains.
2. Physiology itself is not static. Day-to-day variability driven (likely) by recovery status, accumulated load, sleep, nutrition, and normal biological fluctuation means that the "true" CP is not a perfectly stable value.

For these reasons, CP should be considered a (small) transition bandwidth, not a single-point metric. (2) Practically speaking, if testing yields a CP of 350 W (assuming the test quality is solid), it is more realistic to interpret it roughly as a 340–355/360 W range.

This is precisely why, when building a 5/7 training zones model), the suggestion is defining **Zone 4 as ±5/10% around CP**. Doing so ensures that:

- **Zone 3** clearly sits within the *medium/heavy* intensity domain
- **Zone 5** clearly falls within the *high/severe* intensity domain
- **Zone 4** captures the maximum metabolic steady-state transition (MMSS)

Within that Zone 4 bandwidth, you can target the lower, middle, or upper end depending on the goal of the session and how you feel on the day — while still training within the intended metabolic domain.

So how do you test and obtain your CP and W′ values? Performing a series of all-out efforts within the high/severe-intensity domain. To obtain a valid and accurate estimate, the CP testing protocol should follow precise guidelines: number of efforts, duration/intensity of efforts, recovery between efforts, terrain, position on the bike, cadence and pacing.

The power and duration values obtained from these trials should then be plotted, and CP and W′ calculated using each of the three mathematical models described above.

a. The 2 Parameters Hyperbolic Model (CP 2-P Hyp)

b. The 2 Parameters Linear Power-Inverse Time Model (CP 2-1/time)

c. The 2 Parameters Linear Work–Time Model (CP 2-linear)

Image from Leo et al. 2022. (10).

You will likely obtain three different CP values and three different W′ values, each corresponding to one of the three power–duration mathematical models described above. Why different values? While in a virtual perfect dataset all the 3 models would theoretically yield identical CP and W′ estimates, real-world human performance is inherently variable, meaning the mathematical fit is never 100% correct. As a result, each model can produce different parameter estimates (CP and W′) due to unavoidable measurement error and biological variability. So, which CP and W′ should you choose? **You should apply the 'best individual fit' method, that is selecting the CP and W′ values derived from the one of the three models that demonstrates the highest accuracy and precision. (1, 14)** Failing to apply the best individual fit approach can lead to meaningful under- or overestimation of Critical Power (CP). Moreover, if none of the three models provides adequate accuracy and precision, you should consider repeating one or more of the all-out trials. In most cases, this reflects suboptimal pacing or underperformance in at least one effort.

In the next KIW Research Note, we will outline practical guidelines for CP and W′ testing, explain how to implement the best individual fit method in applied settings, and provide access to our new web application that automatically estimates Critical Power and W′ using this approach!

1. Establishing training zones requires knowing two exercise thresholds: the first lactate/ventilatory threshold (LT1/VT1) and the **Maximum Metabolic Steady State (MMSS).**
2. The **MMSS** represents the highest exercise intensity at which a steady state in oxygen consumption and muscle metabolites can still be maintained. It marks the transition between the *heavy/medium* intensity domain and the *high/severe* intensity domain.
3. The **Maximum Lactate Steady State (MLSS)** can provide an accurate and valid estimate of MMSS. However, it is impractical, as its assessment typically requires 4–6 separate 30-minute constant-power trials (at least three), each performed on different days.
4. While calculating the **Second Lactate Threshold (LT2) from an incremental step test** is more practical, there is currently no gold-standard procedure for estimating MLSS — and therefore MMSS — using LT2. This is due to several reasons:
   - There are multiple definitions of LT2, each producing different power outputs. The threshold you identify depends on the method you choose.
   - LT2 results are heavily influenced by the design of the incremental test: step duration, power increase between steps, number of steps, and starting intensity. Even the same LT2 definition (Dmax for example) can yield different values depending on these test parameters. Research has not yet consistently validated any specific test protocol for LT2 as a reliable means of accurately estimating MLSS.
5. **Critical Power (CP), the asymptote of the curve between power output and sustainable duration, represents a practical measure that has been repeatedly shown in scientific studies to accurately estimate the Maximum Metabolic Steady State (MMSS).**
6. To ensure this robust physiological basis, CP testing must follow precise guidelines, and the quality of the estimation can be evaluated using statistical and mathematical indices (the best individual fit method).

We will see point 6 (in a practical and easy way) in the upcoming KIW Research Note!

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Sciences

Founder of Knowledgeiswatt

1. Black MI, Jones AM, Blackwell JR, Bailey SJ, Wylie LJ, McDonagh ST, Thompson C, Kelly J, Sumners P, Mileva KN, Bowtell JL, Vanhatalo A. Muscle metabolic and neuromuscular determinants of fatigue during cycling in different exercise intensity domains. J Appl Physiol (1985). 2017 Mar 1;122(3):446-459.
2. Jones AM, Burnley M, Black MI, Poole DC, Vanhatalo A. The maximal metabolic steady state: redefining the 'gold standard'. Physiol Rep. 2019 May;7(10):e14098. doi: 10.14814/phy2.14098. PMID: 31124324; PMCID: PMC6533178.
3. Sitko S, Artetxe X, Bonnevie-Svendsen M, Galán-Rioja MÁ, Gallo G, Grappe F, Leo P, Mateo M, Mujika I, Sanders D, Seiler S, Zabala M, Valenzuela PL, Viribay A. What Is "Zone 2 Training"?: Experts' Viewpoint on Definition, Training Methods, and Expected Adaptations. Int J Sports Physiol Perform. 2025 Feb 26;20(11):1614-1617. doi: 10.1123/ijspp.2024-0303. PMID: 40010355.
4. Beneke R. Methodological aspects of maximal lactate steady state-implications for performance testing. Eur J Appl Physiol. 2003 Mar;89(1):95-9. doi: 10.1007/s00421-002-0783-1. Epub 2003 Jan 21. PMID: 12627312.
5. Bonaventura JM, Sharpe K, Knight E, Fuller KL, Tanner RK, Gore CJ. Reliability and accuracy of six hand-held blood lactate analysers. J Sports Sci Med. 2015 Mar 1;14(1):203-14. PMID: 25729309; PMCID: PMC4306774.
6. Jamnick NA, Botella J, Pyne DB, Bishop DJ. Manipulating graded exercise test variables affects the validity of the lactate threshold and V˙O2peak. PLoS One. 2018 Jul 30;13(7):e0199794. doi: 10.1371/journal.pone.0199794. PMID: 30059543; PMCID: PMC6066218.
7. Hill, A. V. 1925. The physiological basis of athletic records. *Nature* **116**: 544–548.
8. Burnley M, Vanhatalo A, Poole DC, Jones AM. Blue plaque review series: A.V. Hill, athletic records and the birth of exercise physiology. J Physiol. 2025 Mar;603(6):1361-1374. doi: 10.1113/JP288130. Epub 2025 Feb 23. PMID: 39988844; PMCID: PMC11908475.
9. Poole DC, Burnley M, Vanhatalo A, Rossiter HB, Jones AM. Critical Power: An Important Fatigue Threshold in Exercise Physiology. Med Sci Sports Exerc. 2016 Nov;48(11):2320-2334. doi: 10.1249/MSS.0000000000000939. PMID: 27031742; PMCID: PMC5070974.
10. Leo P, Spragg J, Podlogar T, Lawley JS, Mujika I. Power profiling and the power-duration relationship in cycling: a narrative review. Eur J Appl Physiol. 2022 Feb;122(2):301-316. doi: 10.1007/s00421-021-04833-y. Epub 2021 Oct 27. PMID: 34708276; PMCID: PMC8783871.
11. Jones AM, Vanhatalo A, Burnley M, Morton RH, Poole DC. Critical power: implications for determination of V˙O2max and exercise tolerance. Med Sci Sports Exerc. 2010 Oct;42(10):1876-90. doi: 10.1249/MSS.0b013e3181d9cf7f. PMID: 20195180.
12. Poole DC, Ward SA, Gardner GW, Whipp BJ. Metabolic and respiratory profile of the upper limit for prolonged exercise in man. Ergonomics. 1988 Sep;31(9):1265-79. doi: 10.1080/00140138808966766. PMID: 3191904.
13. Jones AM, Wilkerson DP, DiMenna F, Fulford J, Poole DC. Muscle metabolic responses to exercise above and below the "critical power" assessed using 31P-MRS. Am J Physiol Regul Integr Comp Physiol. 2008 Feb;294(2):R585-93. doi: 10.1152/ajpregu.00731.2007. Epub 2007 Dec 5. PMID: 18056980.
14. Black MI, Jones AM, Bailey SJ, Vanhatalo A. Self-pacing increases critical power and improves performance during severe-intensity exercise. Appl Physiol Nutr Metab. 2015 Jul;40(7):662-70. doi: 10.1139/apnm-2014-0442. Epub 2015 Mar 11. PMID: 26088158.